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Duration	2 mins 57 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Correct

Mark 1.00 out of 1.00

What is the time complexity of linear search in the worst case?

a. O(log n)
b. O(n log n)
c. O(1)
d. O👎 ✓

Question **2**

Correct

Mark 1.00 out of 1.00

In linear search, if the target element is not found in the list, what is the result?

a. The search is considered unsuccessful ✓
b. An error is raised
c. The last element is returned
d. The first element is returned

Question **3**

Correct

Mark 1.00 out of 1.00

What is the advantage of binary search over linear search?

a. Binary search has a lower time complexity on large, sorted lists ✓
b. Binary search works on unsorted lists
c. Binary search can find multiple instances of the target element
d. Binary search does not require dividing the list

Question **4**

Correct

Mark 1.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

a. There must be a mechanism to delete and/or insert elements in the list ✓

b. Number values should only be present

Number values should only be present

c. There should be direct access to the middle element in any sublist

d. The list must be sorted

Question **5**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

a. Linear search requires the list to be sorted.

b. Linear search divides the list into halves.

c. Linear search is more efficient than binary search on large lists.

d. Linear search can be applied to both sorted and unsorted lists. ✓

Question **6**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

a. n ✓

b. n-1

c. n/2

d. log n

Question **7**

Correct

Mark 1.00 out of 1.00

Which of the following scenarios is best suited for applying binary search?

- a. When the list contains duplicate elements
- b. When the list is very small
- c. When the list is unsorted
- d. When the list is sorted ✓

Question **8**

Correct

Mark 1.00 out of 1.00

What type of search would be most appropriate for finding an element in a list that is frequently updated?

- a. Binary search
- b. Hash search
- c. Interpolation search
- d. Linear search ✓

Question **9**

Correct

Mark 1.00 out of 1.00

What happens in a binary search if the list has an even number of elements?

- a. The lower middle element is chosen as the middle element
- b. The search stops
- c. The middle element is chosen randomly
- d. The higher middle element is chosen as the middle element ✓

Question **10**

Correct

Mark 1.00 out of 1.00

Which of the following is a type of searching method?

- a. Linear search ✓
- b. Merge search
- c. Quick search
- d. Bubble search

Question **11**

Correct

Mark 1.00 out of 1.00

In _____ checks the elements of a list, one at a time, without skipping any element.

- a. Linear search ✓
- b. Binary search
- c. Hash search
- d. Both (1) & (3)

Question **12**

Correct

Mark 1.00 out of 1.00

Which of the following is a conventional searching technique?

- a. Hashing
- b. Binary search
- c. Dynamic search
- d. Linear search ✓

Question **13**

Correct

Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

- a. The search continues from the beginning
- b. The search stops
- c. The list is sorted
- d. The search continues in the left or right sublist ✓

Question **14**

Correct

Mark 1.00 out of 1.00

In linear search, how is the element searched?

- a. By comparing each element in the list sequentially ✓
- b. By sorting the list first
- c. By using a hash function
- d. By dividing the list into halves

Question **15**

Correct

Mark 1.00 out of 1.00

_____ search takes a sorted/ordered list and divides it in the middle.

- a. Linear
- b. Hash
- c. Binary ✓
- d. Both (1) & (3)